2.7 Apparent Weight

Learning Goals (Students will be able too….)

- Understand and define apparent weight.
- Be able to solve problems where you need to calculate for apparent weight.

Additional Resources

Reading: Weightlessness in Orbit – Physics Classroom.

Accelerating Frames of References

- **Recall:** Newton's first law is also known as the law of Inertia, that discusses inertial frames of references. An object at rest remains at rest and an object in motion continues in motion with constant velocity unless it is acted on by an external net force.
- We are now going to be exploring what happens when we have an **object at rest** but our **frame of reference is accelerating** such as an elevator.

Normal Force – Thinking About an Elevator Ride

Elevators do **6** things:

1) Accelerate upwards from the bottom.
2) Move at a constant speed upwards.
3) Slow down as they reach the top.
4) Accelerate Downwards from the top.
5) Move at a constant speed downward.
6) Slow down as they reach the bottom.

Activity: Sort these motions according to how you would feel on the elevator at the time.

Feel heavier than usual	Feel the same weight as usual	Feel lighter than usual
1, 6	2, 5 constant speed	3, 4

1) How do you think the normal force compares to the force of gravity in each of these situations?

Heavier : $F_N > F_g$

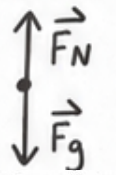
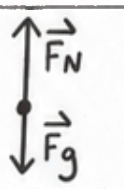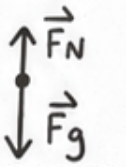
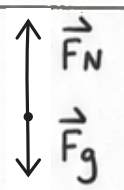
Same : $F_N = F_g$

Lighter : $F_N < F_g$

2) Fill in the table below. Consider the direction of acceleration FIRST for each situation and make sure your net force has the same direction.

Videos/ Simulations:
https://www.youtube.com/watch?v=hPEx3gxtPK4

https://javalab.org/en/elevator_en/

	Direction of Acceleration	FBD
moving at constant speed upwards or downwards	$\vec{a} = 0$	$F_N = F_g$ 
accelerating upwards from the bottom,	$\vec{a} = \uparrow$	 $F_N > F_g$
slowing down as it reaches the top	$\vec{a} = \downarrow$	 $F_N < F_g$
accelerating downwards from the top	$\vec{a} = \downarrow$	 $F_N < F_g$
slowing down as it reaches the bottom	$\vec{a} = \uparrow$	 $F_N > F_g$

3) Based on your FBD's, circle the correct statement:
   a. I feel heavier when the force of gravity on me is greater
   b. I feel heavier when the normal force on me is greater

4) Explain **why** the normal force affects how heavy we feel.

Because the normal force is the force exerted by the surface supporting our body, and we perceive this force as our apparent weight.

5) Circle the best statement:

a. the normal force is a measure of the force of gravity
b. the normal force is a measure of the applied force
c. the normal force is a measure of how much two surfaces are being pressed together

Apparent Weight

- **Apparent Weight:** Is defined by how heavy WE feel. It is fictitious in the sense of it is how we perceive our weight, and it is calculated by the normal force on our body. We can calculate our apparent mass (m_{app}) according to *apparent weight* = $N = m_{app}g$. Just realize that the mass doesn't actually change.

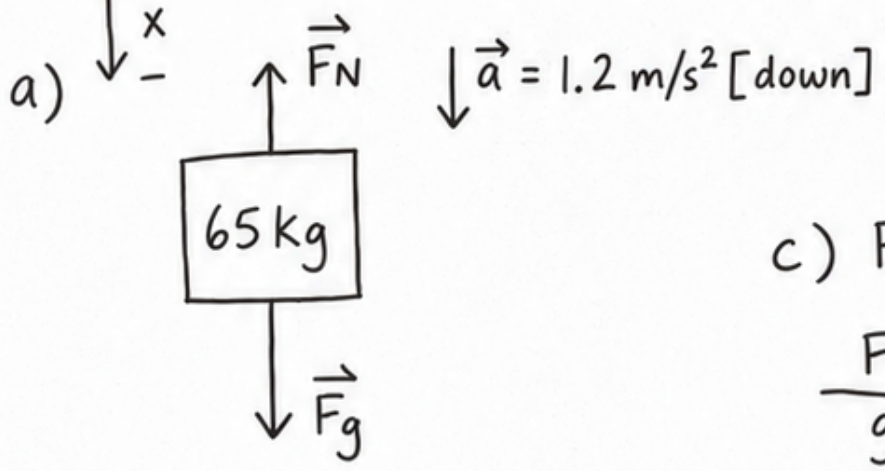
Class Example

1) An elevator is moving upwards, accelerating at 1.2 m/s^2 [down]. A student of mass 65 kg stands on a bathroom scale in this elevator. What is the reading on the scale (in kg)?

To solve the problem, break it down accordingly:

a) draw an fbd
b) determine the normal force on the student [560 kg]
c) determine the apparent mass [57 kg]

a)



b) $F_{net} = F_N - F_g$

$F_{net} = -ma$ → mg

$-ma = F_N - F_g$

$-65 \times 1.2 = F_N - (65 \times 9.81)$

$-65 \times 1.2 + (65 \times 9.81) = F_N$

$F_N = 559.65\ N$

$F_N = 560\ N$

c) $F_N = m_{app}\, g$

$\frac{F_N}{g} = m_{app}$

$m_{app} = \frac{559.65}{9.81}$

$m_{app} = 57.05\ kg$

$m_{app} = 57\ kg$